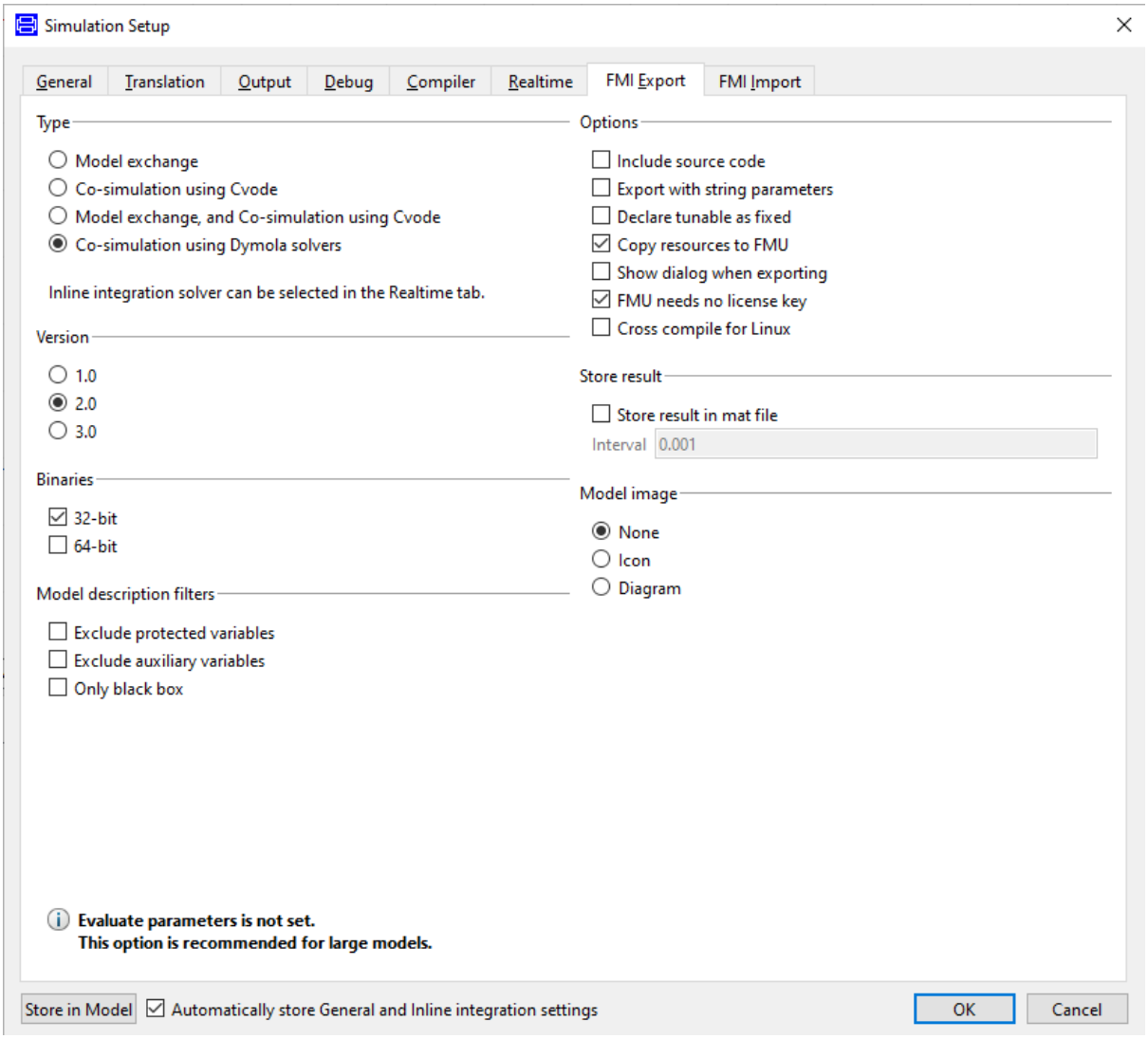
Dymola Export to NI VeriStand – Steps

The following steps document the complete workflow for exporting the ThreeZoneBuilding_PHiL model from Dymola as an FMU, compiling the custom NIRT wrapper DLL, and loading it into NI VeriStand 2018.

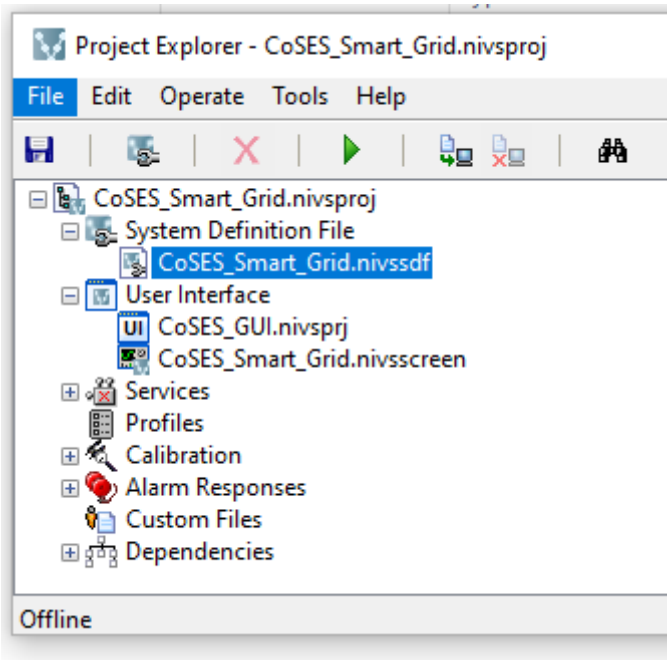
1. Open the Dymola software, load and locate the model file (ThreeZoneBuilding_PHiL_optimized_withWeather) for simulation
2. Enable simulation and under Simulation tab – open Setup tab as attached.



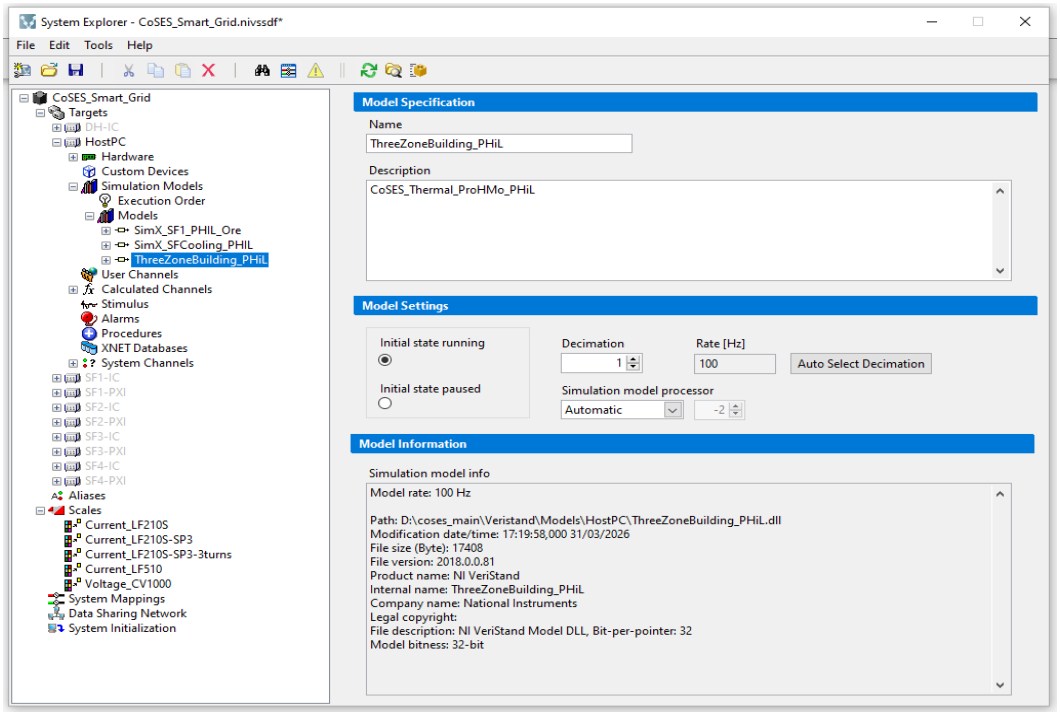
3. Under Compiler tab, make sure to add custom options as /arch:IA32. Also, please note that the C compiler checked is already installed on the laboratory PC.
4. FMI Export tab, make sure the following and click ok:
   - Type: **Co-Simulation using Dymola solvers**
   - Algorithm: DASSL
   - Version: 2.0
   - Binaries: 32-bit
   - Options: Copy resources to FMU; FMU needs to license key.
5. Click the dropdown arrow below Translate and select FMU

6. Soon as output log shows “Finished”, navigate to the Dymola current working directory, look for latest .fmu extension file. Rename them into .zip format and extract them.
7. Once the file opens up, navigate to binaries/win32 and you can find the .dll file, rename the file to tzb_fmu.dll
8. Copy tzb_fmu.dll to D:\coses_main\Veristand\Models\HostPC. Also ensure the extracted FMU resources folder (ThreeZoneBuilding_PHiL_optimized_withWeather) is present in the same directory. This path is hardcoded in the wrapper as the FMU resource URI; the runtime will fail to instantiate the model if the resources folder is missing or placed elsewhere.
9. Also open modelDescription.xml file, look for the “ScalarVariable name” you want as inputs/ outputs and copy the “valueReference” number for fmu_veristand_wrapper.c file.
10. Navigate to “x86 Native Tools Command Prompt for VS 2019” and run the following commands
    - Navigate to directory: *cd /D D:\coses_main\Veristand\Models\HostPC*
    - Look for c wrapper file: *dir fmu_veristand_wrapper.c*
    - fmu_veristand_wrapper.c is a manually authored custom C source file specific to this project. It should already be present in the HostPC directory. If it is missing, copy it from the project repository (CoSES_Thermal_ProHMo_PHiL) before proceeding.
    - Edit the file and enter IN_NAMES[], OUT_NAMES[], vr_in[] and vr_out[] arrays with the corresponding ScalarVariable name and valueReference respectively as obtained in step 10.
11. Once you see listed file with size in bytes, compile the following: *cl /LD /arch:IA32 fmu_veristand_wrapper.c /FeThreeZoneBuilding_PHiL.dll kernel32.lib msvcrt.lib /link /NODEFAULTLIB:LIBCMT /MACHINE:X86*
12. If compilation succeeds but the model fails to load or outputs remain zero, verify the FMU_GUID value in fmu_veristand_wrapper.c. Open modelDescription.xml (from the extracted FMU) and locate the guid= attribute in the root fmiModelDescription element. The value must match the FMU_GUID #define in the wrapper exactly (e.g., {0c4fc5dd-8578-4e04-84d8-dd2e48eec986}). If they differ, update the #define and recompile.
13. Three precautions address NIRT runtime exceptions. First, the solver in Dymola must be set to DASSL before FMU export; CVODE causes a hard crash on the first DoStep call in the VeriStand NIRT runtime. Second, the wrapper skips DoStep at t=0 to avoid initialization-phase faults. Third, a structured exception handler (SEH __try/__except) in NIRT_Schedule catches any remaining DoStep crash at runtime and automatically reinitializes the FMU component, allowing the run to continue. If zone temperatures read as 0°C on startup, verify that the Modelica energy dynamics parameter is set to FixedInitial (not SteadyStateInitial) before re-exporting the FMU.
14. Once compile is done, you would be able to see a new .dll file (in the name mentioned in c wrapper file)

15. Open NI VeriStand 2018, open the recent CoSES_Smart_Grid Project Explorer, dropdown the System Definition File and open CoSES_Smart_Grid.nivssdf

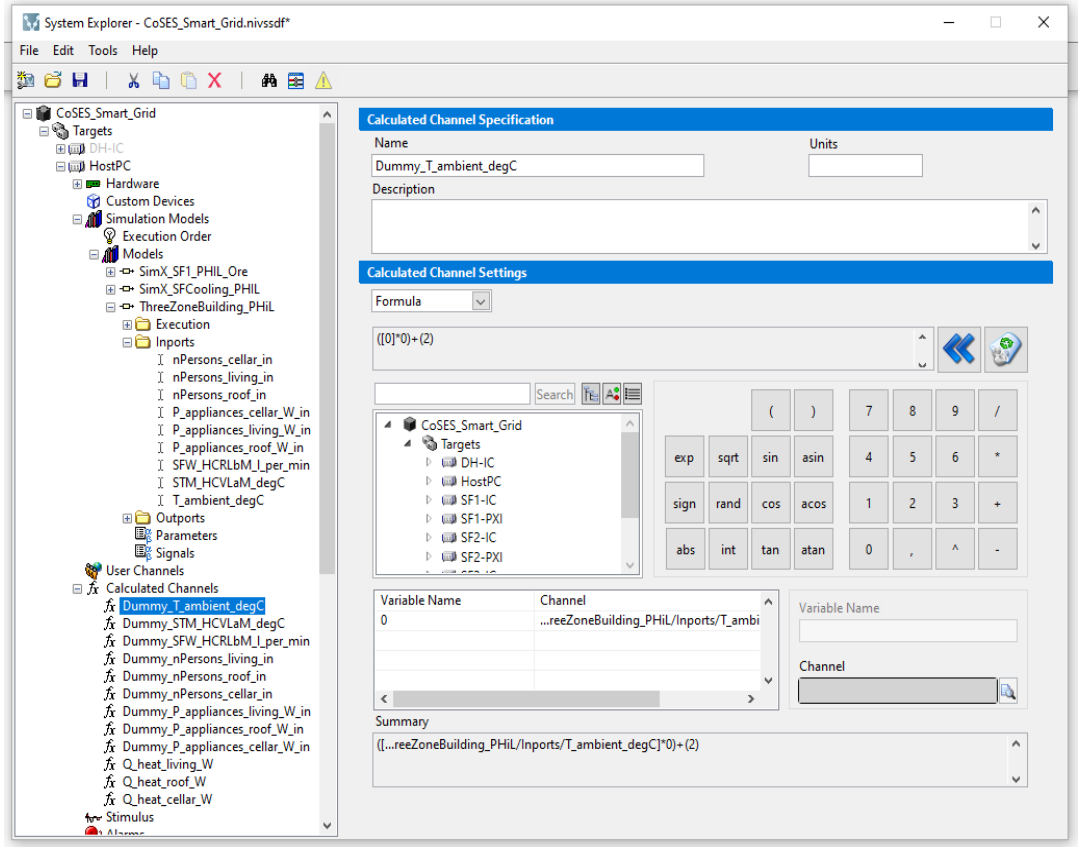


16. Navigate to CoSES_Smart_Grid/Targets/HostPC/Simulation Models and right click and "Add Simulation Model"
17. Load the new .dll file after the compile – from HostPC directory. Once it loads click ok and it looks like the attached.

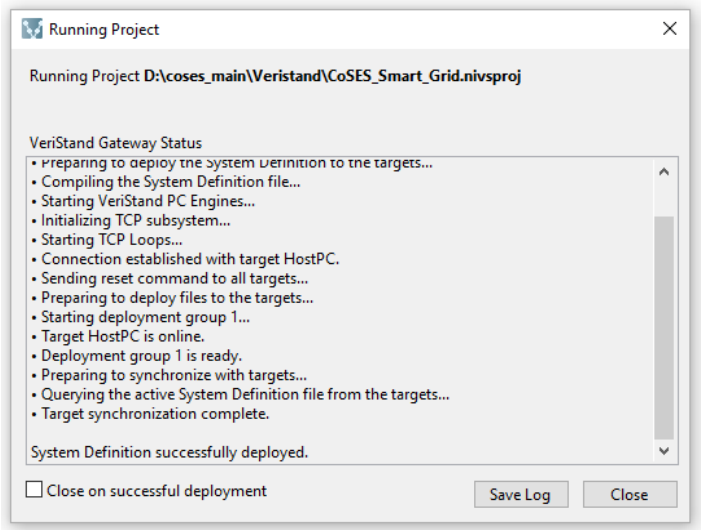


18. Check if the Inports and Outports are same as you mentioned in the custom c wrapper file.

19. Add the inputs manually. Moreover, look out for Calculated Channels and create dummy values for each inputs (matching value of inports) in the format as attached.



20. Save and close the System Explorer and then Deploy the interface.
21. Wait for the log to update and once you receive the below log output as “System Definition successfully deployed” close the log and you can open CoSES_Smart_Grid.nivsscreen (found under User Interface button)



22. Navigate to File - Screen - Add Screen and add all the necessary workspace controls under the edit mode.

Inputs
Temperature / Flow
Appliance Load
Occupancy
T_ambient_degC
2,0
P_appliances_living_W
500,0
nPersons_living_in
3,0
STM_HCVLaM_degC
50,0
P_appliances_roof_W_i
200,0
nPersons_roof_in
1,0
SFW_HCRLbM_l_per_m
12,0
P_appliances_cellar_W
100,0
nPersons_cellar_in
1,0
HostPC/
32,0
Model running...
Outputs
Temperature
Flow
Valve Opening
STM_HCRL_Set_degC
55,5
SFW_HCRLbM_Set_l_p
9,1
valve_living_opening
0,1
T_roomls_degC
20,7
Q_flow_living_W
323,2
valve_cellar_opening
0,3
T_roofls_degC
19,1
Q_flow_roof_W
323,2
valve_roof_opening
0,3
T_cellarls_degC
18,1
Q_flow_cellar_W
323,2